

GED Science Practice Test 14

PART A — QUESTIONS

GED-SCI14-001

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A researcher studies whether hydration affects short-term attention. Group A drinks 500 mL of water one hour before a computer attention test. Group B drinks no water during that hour. Both groups eat the same breakfast and take the same test in the same room.

Group | Average correct responses out of 50

A | 43

B | 36

Which conclusion is best supported by the data?

A) Drinking water before the test was associated with more correct responses.

B) Drinking water before the test was associated with fewer correct responses.

C) Hydration had no possible relationship to attention.

D) Both groups had identical attention-test results.

GED-SCI14-002

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

Torque can be calculated as $\text{force} \times \text{distance from the pivot}$.

Lever | Force (N) | Distance from pivot (m)

A | 20 | 0.5

B | 15 | 1.0

C | 12 | 1.5

D | 8 | 2.0

Which lever setup produces the greatest torque?

A) Lever A

B) Lever B

C) Lever C

D) Lever D

GED-SCI14-003

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

Lithium is used in many rechargeable batteries. Mining lithium can provide material for energy-

storage technology, but mining may also use large amounts of water and disturb habitats. Recycling batteries can recover some lithium and other metals.

Which statement best describes one environmental benefit of battery recycling?

- A) It can reduce the need to mine new raw materials.
- B) It creates unlimited lithium from air.
- C) It stops all use of batteries immediately.
- D) It changes lithium into a renewable food source.

GED-SCI14-004

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Blood calcium is controlled partly by two hormones. Parathyroid hormone can raise blood calcium when levels are too low. Calcitonin can help lower blood calcium when levels are too high. These responses help keep calcium within a narrow range needed for nerve and muscle function.

Which idea is best illustrated by this process?

- A) Hormonal feedback helps maintain homeostasis.
- B) Calcium levels never change in the blood.
- C) Hormones prevent nerves and muscles from working.
- D) Blood calcium is controlled only by the lungs.

GED-SCI14-005

Section: Science

Type: numeric_entry

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A person pushes down on a wrench with a force of 90 N. The force is applied 0.5 m from the bolt.

$\text{Torque} = \text{force} \times \text{distance from pivot}$

What torque is produced in N·m?

Type your answer as a whole number.

GED-SCI14-006

Section: Science

Type: multi_select

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A student tests how axle position affects how far a small cart rolls after a push. The same cart body and the same starting push are used in every trial. Only the axle position is changed.

Which TWO factors should be kept constant?

- A) Cart body

- B) Starting push
- C) Axle position
- D) Distance rolled
- E) Final measured result

GED-SCI14-007

Section: Science

Type: dropdown

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Some traits are influenced by gene expression, which means whether a gene is actively used to make a product. Environmental factors can sometimes affect gene expression without changing the DNA sequence. This kind of regulation affects [D1] and does not necessarily change [D2].

Dropdowns:

D1) how strongly a gene is expressed | the orbit of a moon | rock hardness

D2) the DNA sequence itself | sunlight direction only | soil grain size

GED-SCI14-008

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student carefully combines a weak acid and a weak base in a beaker. The temperature rises slightly, and the final solution is closer to neutral than either starting solution. The student identifies the process as neutralization.

Which statement best describes neutralization?

- A) An acid and a base react to form a more neutral solution.
- B) A solid changes directly into a gas without energy transfer.
- C) A magnet separates iron from sand.
- D) A chromosome copies itself before cell division.

GED-SCI14-009

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

Astronomers observe that light from a distant galaxy is shifted toward longer, redder wavelengths. This redshift is evidence that the galaxy is moving away from Earth. Observations of many galaxies show a similar pattern.

Which conclusion is best supported by this evidence?

- A) Many distant galaxies are moving away from Earth.
- B) Galaxies are made only of liquid water.
- C) Redshift means stars have stopped producing light.
- D) Earth is the only object in the universe that moves.

GED-SCI14-010

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A doctor measures kidney filtration rate in four patients.

Patient | Estimated filtration rate (mL/min)

A | 96

B | 72

C | 48

D | 88

A lower filtration rate may indicate reduced kidney function.

Which patient has the lowest estimated filtration rate?

A) Patient A

B) Patient B

C) Patient C

D) Patient D

GED-SCI14-011

Section: Science

Type: drag_drop

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Match each target with the correct tile.

Tiles:

A) Number of gear teeth on the driver gear

B) Rotation speed of the output gear

C) Same driven gear used in every trial

D) Color of the gear stand

Targets:

T1) Independent variable

T2) Dependent variable

T3) Controlled variable

GED-SCI14-012

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Bone marrow contains stem cells that can produce different types of blood cells. Some cells develop into red blood cells that carry oxygen. Others develop into white blood cells involved in immune defense or platelets involved in clotting.

Which statement best describes one function of bone marrow?

A) It produces cells that become blood cells.

- B) It bends light through a glass lens.
- C) It filters groundwater through sand.
- D) It forms lava from cooled magma.

GED-SCI14-013

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

Two sound waves meet in the same space. In one case, the high points of the waves line up and the sound becomes louder. In another case, a high point lines up with a low point and the sound becomes quieter.

Which wave behavior is described?

- A) Interference
- B) Digestion
- C) Sublimation
- D) Plate collision

GED-SCI14-014

Section: Science

Type: numeric_entry

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A city reduces annual coal use from 50,000 tons to 38,000 tons after switching part of its energy supply to renewable sources.

What is the percent decrease in coal use?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI14-015

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A researcher tests whether a new meal plan affects cholesterol level. One group follows the meal plan for 10 weeks. A comparison group continues its usual diet. Both groups have cholesterol measured at the beginning and end of the study.

Why should cholesterol be measured before and after the study?

- A) To compare changes in cholesterol over time for each group
- B) To make cholesterol impossible to detect
- C) To guarantee both groups have the same final result
- D) To avoid using any data from the comparison group

GED-SCI14-016

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student compares how different blade pitches affect the lift produced by a small fan blade. The same motor and same blade material are used for every trial. A sensor measures lift force.

What is the dependent variable?

- A) Lift force measured by the sensor
- B) Blade pitch
- C) Same motor
- D) Same blade material

GED-SCI14-017

Section: Science

Type: dropdown

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A fixed pulley changes the direction of an input force. A movable pulley can reduce the input force needed to lift a load. In this description, the fixed pulley mainly changes [D1], while the movable pulley can provide [D2].

Dropdowns:

D1) force direction | genetic variation | blood glucose

D2) mechanical advantage | photosynthesis | chromosome separation

GED-SCI14-018

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A trainer measures muscle fatigue by recording how long participants can hold a light weight with the arm extended.

Participant | Hold time (seconds)

- A | 42
- B | 67
- C | 51
- D | 74

Which participant held the weight the longest?

- A) Participant A
- B) Participant B
- C) Participant C
- D) Participant D

GED-SCI14-019

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares the output of different gear pairs.

Gear pair | Input rotations | Output rotations

A | 10 | 20

B | 10 | 15

C | 10 | 10

D | 10 | 5

Which gear pair produces the greatest increase in rotation number?

A) Gear pair A

B) Gear pair B

C) Gear pair C

D) Gear pair D

GED-SCI14-020

Section: Science

Type: multi_select

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

Select TWO answers.

A coastal region tracks mangrove coverage and estimated storm-damage cost.

Year | Mangrove coverage (hectares) | Estimated storm-damage cost (million \$)

1 | 1,200 | 86

2 | 1,450 | 74

3 | 1,700 | 63

4 | 1,950 | 55

Which TWO statements are best supported by the data?

A) Mangrove coverage increased from Year 1 to Year 4.

B) Estimated storm-damage cost decreased from Year 1 to Year 4.

C) Mangrove coverage decreased every year.

D) Storm-damage cost was highest in Year 4.

E) Both measurements stayed exactly the same.

GED-SCI14-021

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

During meiosis, chromosome pairs separate so reproductive cells receive one chromosome from each pair. If chromosomes fail to separate correctly, a reproductive cell may receive too many or too few chromosomes.

Which result can be caused by this kind of error?

A) An abnormal chromosome number in an offspring

B) A change in mineral hardness

- C) A decrease in the Moon's orbital period
- D) Formation of igneous rock from lava

GED-SCI14-022

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

Dry ice is solid carbon dioxide. At room temperature, dry ice changes directly from a solid to a gas without becoming liquid first.

Which phase change is described?

- A) Sublimation
- B) Freezing
- C) Condensation
- D) Cementation

GED-SCI14-023

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A metamorphic rock forms near a magma intrusion. The original rock does not melt, but heat from the nearby magma changes its minerals and texture. Pressure is not the main factor in this case.

Which process is described?

- A) Contact metamorphism
- B) River deposition
- C) Solar fusion
- D) Wind transport only

GED-SCI14-024

Section: Science

Type: numeric_entry

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A hospital reviews 420 patient records. Of these records, 63 show a documented medication allergy.

What percent of the records show a documented medication allergy?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI14-025

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student tests how the thickness of protective padding affects peak impact force. The same mass is dropped from the same height onto padding samples of different thicknesses. A force sensor records the peak impact force.

Which factor is the independent variable?

- A) Padding thickness
- B) Peak impact force
- C) Dropped mass
- D) Drop height

GED-SCI14-026

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Peroxisomes are organelles that contain enzymes used to break down certain fatty acids and detoxify harmful substances such as hydrogen peroxide. These reactions help protect the cell from chemical damage.

Which statement best describes peroxisomes?

- A) They help break down harmful substances inside cells.
- B) They move tectonic plates under continents.
- C) They produce sunlight during photosynthesis.
- D) They carry oxygen through arteries.

GED-SCI14-027

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student records the height reached by the same ball after it is launched with different stored spring energies.

Stored spring energy (J): 2, 4, 6, 8

Maximum height (m): 0.4, 0.8, 1.2, 1.6

Which statement is best supported by the data?

- A) Greater stored spring energy was associated with greater maximum height.
- B) Greater stored spring energy was associated with lower maximum height.
- C) Maximum height stayed the same for every energy value.
- D) Stored spring energy had no relationship to ball height.

GED-SCI14-028

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares the rotation periods of four planets.

Planet | Rotation period (Earth hours)

- A | 10
- B | 24
- C | 58
- D | 243

Which planet has the shortest rotation period?

- A) Planet A
- B) Planet B
- C) Planet C
- D) Planet D

GED-SCI14-029

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A doctor records blood glucose level after a patient receives insulin.

Time after insulin (min): 0, 30, 60, 90

Blood glucose (mg/dL): 190, 154, 118, 96

Which statement is best supported by the data?

- A) Blood glucose decreased after insulin was given.
- B) Blood glucose increased after insulin was given.
- C) Blood glucose stayed exactly the same.
- D) Insulin made blood glucose disappear instantly.

GED-SCI14-030

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student places a warm metal block on a cooler ceramic tile. After several minutes, the metal block cools and the ceramic tile warms.

Which statement best describes the energy transfer?

- A) Thermal energy transfers from the warmer metal block to the cooler tile.
- B) Cold energy moves from the tile into the metal block.
- C) Thermal energy cannot move between solids.
- D) The tile produces gravity to cool the metal.

GED-SCI14-031

Section: Science

Type: multi_select

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A researcher studies whether a new coordination exercise improves balance in people recovering from a mild nerve injury. One group completes the exercise program, and another group receives usual care. Balance is measured after six weeks.

Which TWO choices would improve the investigation?

- A) Use the same balance-measurement method for both groups.
- B) Include enough participants in both groups.
- C) Record only the results from the exercise group.
- D) Give one group a longer testing period.
- E) Change the balance test for each participant.

GED-SCI14-032

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student measures the time for different objects to complete one full vibration.

Object | Time for 1 vibration (s)

A | 0.25

B | 0.50

C | 1.00

D | 2.00

Frequency = $1 \div \text{period}$

Which object has the highest frequency?

- A) Object A
- B) Object B
- C) Object C
- D) Object D

GED-SCI14-033

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

After a major flood, a wetland area remains covered with nutrient-rich sediment. In the following years, fast-growing plants spread first, followed by shrubs and young trees as the habitat changes.

Which ecological process is described?

- A) Secondary succession
- B) Electrical induction
- C) Lunar eclipse
- D) Acid-base neutralization

GED-SCI14-034

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A person inherits a changed copy of a gene involved in blood clotting. The change can reduce the amount of working clotting protein, making bleeding last longer after an injury.

Which statement best explains the effect of the gene change?

- A) A gene change can affect a protein involved in blood clotting.
- B) A gene change always improves clotting.
- C) Clotting proteins are made only by rocks.
- D) Inherited genes cannot affect body functions.

PART B — ANSWER KEY + EXPLANATIONS

GED-SCI14-001 — Correct: A — Correct Answer: Drinking water before the test was associated with more correct responses. — Explanation: Group A drank water and had a higher average score than Group B.

GED-SCI14-002 — Correct: C — Correct Answer: Lever C — Explanation: Torque = force $\times$ distance. Lever C produces $12\text{ N} \times 1.5\text{ m} = 18\text{ N}\cdot\text{m}$, the greatest value listed.

GED-SCI14-003 — Correct: A — Correct Answer: It can reduce the need to mine new raw materials. — Explanation: Recycling can recover useful metals and reduce demand for new mining.

GED-SCI14-004 — Correct: A — Correct Answer: Hormonal feedback helps maintain homeostasis. — Explanation: Hormones that raise or lower blood calcium help keep internal conditions stable.

GED-SCI14-005 — Correct: 45 — Tolerance: exact — Correct Answer: 45 — Explanation: Torque = $90\text{ N} \times 0.5\text{ m} = 45\text{ N}\cdot\text{m}$.

GED-SCI14-006 — Correct: A,B — Correct Answer: Cart body | Starting push — Explanation: The cart body and starting push should stay the same so axle position is the tested variable.

GED-SCI14-007 — Correct: D1=how strongly a gene is expressed; D2=the DNA sequence itself — Correct Answer: D1=how strongly a gene is expressed; D2=the DNA sequence itself — Explanation: Environmental factors may influence gene expression without changing the DNA sequence.

GED-SCI14-008 — Correct: A — Correct Answer: An acid and a base react to form a more neutral solution. — Explanation: Neutralization occurs when an acid and a base react, producing a solution closer to neutral.

GED-SCI14-009 — Correct: A — Correct Answer: Many distant galaxies are moving away from Earth. — Explanation: Redshift toward longer wavelengths is evidence that distant galaxies are moving away.

GED-SCI14-010 — Correct: C — Correct Answer: Patient C — Explanation: Patient C has the lowest listed filtration rate, 48 mL/min.

GED-SCI14-011 — Correct: T1=A; T2=B; T3=C — Correct Answer: T1=Number of gear teeth on the driver gear; T2=Rotation speed of the output gear; T3=Same driven gear used in every trial — Explanation: The driver gear teeth are changed, output rotation speed is measured, and the driven gear is controlled.

GED-SCI14-012 — Correct: A — Correct Answer: It produces cells that become blood cells. — Explanation: Bone marrow contains stem cells that produce blood cell types.

GED-SCI14-013 — Correct: A — Correct Answer: Interference — Explanation: Waves can combine to make a sound louder or quieter depending on how their high and low points line up.

GED-SCI14-014 — Correct: 24 — Tolerance: exact — Correct Answer: 24 — Explanation: Coal use decreased by 12,000 tons. $12,000 \div 50,000 \times 100 = 24\%$.

GED-SCI14-015 — Correct: A — Correct Answer: To compare changes in cholesterol over time for each group — Explanation: Measuring before and after allows researchers to compare how cholesterol changed.

GED-SCI14-016 — Correct: A — Correct Answer: Lift force measured by the sensor — Explanation: The dependent variable is the measured outcome after changing blade pitch.

GED-SCI14-017 — Correct: D1=force direction; D2=mechanical advantage — Correct Answer: D1=force direction; D2=mechanical advantage — Explanation: A fixed pulley changes force direction, while a movable pulley can reduce input force.

GED-SCI14-018 — Correct: D — Correct Answer: Participant D — Explanation: Participant D held the weight the longest, 74 seconds.

GED-SCI14-019 — Correct: A — Correct Answer: Gear pair A — Explanation: Gear pair A produces 20 output rotations from 10 input rotations, the greatest increase.

GED-SCI14-020 — Correct: A,B — Correct Answer: Mangrove coverage increased from Year 1 to Year 4. | Estimated storm-damage cost decreased from Year 1 to Year 4. — Explanation: Mangrove coverage increased from 1,200 to 1,950 hectares, while estimated damage cost decreased from \$86 million to \$55 million.

GED-SCI14-021 — Correct: A — Correct Answer: An abnormal chromosome number in an offspring — Explanation: Errors in chromosome separation can produce reproductive cells with too many or too few chromosomes.

GED-SCI14-022 — Correct: A — Correct Answer: Sublimation — Explanation: Sublimation is the change from solid directly to gas.

GED-SCI14-023 — Correct: A — Correct Answer: Contact metamorphism — Explanation: Contact metamorphism occurs when heat from nearby magma changes rock without melting it.

GED-SCI14-024 — Correct: 15 — Tolerance: exact — Correct Answer: 15 — Explanation: $63 \div 420 \times 100 = 15\%$.

GED-SCI14-025 — Correct: A — Correct Answer: Padding thickness — Explanation: The independent variable is the factor intentionally changed by the student.

GED-SCI14-026 — Correct: A — Correct Answer: They help break down harmful substances inside cells. — Explanation: Peroxisomes contain enzymes that detoxify certain harmful substances.

GED-SCI14-027 — Correct: A — Correct Answer: Greater stored spring energy was associated with greater maximum height. — Explanation: Maximum height increased from 0.4 m to 1.6 m as stored spring energy increased.

GED-SCI14-028 — Correct: A — Correct Answer: Planet A — Explanation: Planet A has the shortest rotation period, 10 Earth hours.

GED-SCI14-029 — Correct: A — Correct Answer: Blood glucose decreased after insulin was given. — Explanation: Blood glucose decreased from 190 mg/dL to 96 mg/dL over 90 minutes.

GED-SCI14-030 — Correct: A — Correct Answer: Thermal energy transfers from the warmer metal block to the cooler tile. — Explanation: Thermal energy naturally moves from a warmer object to a cooler object.

GED-SCI14-031 — Correct: A,B — Correct Answer: Use the same balance-measurement method for both groups. | Include enough participants in both groups. — Explanation: Consistent measurement and adequate sample size improve reliability.

GED-SCI14-032 — Correct: A — Correct Answer: Object A — Explanation: Frequency = $1 \div \text{period}$. Object A has the shortest period, so it has the highest frequency.

GED-SCI14-033 — Correct: A — Correct Answer: Secondary succession — Explanation: Secondary succession can occur after a disturbance when soil or sediment remains.

GED-SCI14-034 — Correct: A — Correct Answer: A gene change can affect a protein involved in blood clotting. — Explanation: Genes can affect proteins, including proteins involved in blood clotting.